

EAN-Detection Modes

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 **CAUTION:** Alerts to a condition or improper procedure that can result in equipment or component damage if not avoided.

 **IMPORTANT:** Identifies information that is critical to successful setup, configuration, or operation of the system.

 Provides supplementary information that aids in the use or understanding of the system or subject matter but is not critical to operation.



1 Overview

This document provides information and guidelines on the parameters that primarily affect the detection modes. This document will focus on the settings that will have the most impact on detection performance, and those that will need adjustment beyond their default settings.

To understand how the parameter settings impact performance, use this document during evaluation and testing.

1.1 Additional Support Documentation

The [Panel Plus](#) User Guide is accessible from the Help menu within the application.

The Interface Command and Control (IDD) describes the native communications protocol.

Additional Engineering Application Notes (EANs) and ICDs are available on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

1.2 Sightline Software Requirements

IMPORTANT: The Panel Plus software version should match the firmware version running on the OEM. Firmware and Panel Plus software versions are available on the [Software Releases](#) page.

Table 1: Sightline OEM Software Requirements

OEMs	Initial Software Release	Provisions
4110-OEM	3.11.x	--
1710-OEM	3.9.x	--
4100-OEM	3.7.x	Three channel processing requires 3.9.x and above.
1750-OEM	3.6.x	Dual channel processing and Snap All Cams requires 3.7.x and above.
4000-OEM	3.00.xx	AI-Assist and AI-Detect are not supported. Blending requires 3.3.x and above.
Legacy OEMs are supported up to 3.5.x		
1500-OEM	2.20.xx	--
3000-OEM	2.22.xx	Rev B6 requires updated firmware to accommodate the change in memory.

1.3 Software Feature Version Requirements

Software feature availability is determined by software version. Refer to the table below for the minimum software version that supports each feature described in this document. For questions or upgrades, contact [Sales](#).

Table 2: Software Feature Version Requirements

Detection Modes	Initial Software Release
Base Detection Modes	
Anomaly	2.22.xx
Blob	2.24.xx
Gas	2.25.xx - 3.2.x (updated in 3.3.x)
Radiometric	2.23.xx
Staring	2.22.xx
Vehicle	2.22.xx
Person	3.4.xx
Advanced Detection Modes (Base Detection is a pre-requisite for Advanced Detection modes)	
Aerial	2.22.xx
Drone	2.24.xx (as part of base) 2.25.xx (moved to advanced)
Drone w/ Classifier	2.25.xx
Maritime (beta)	2.23.xx
Motion	3.11.xx



(Software Feature Version Requirements table continued)

Detection Modes	Initial Software Release
Gas Detection Mode (Stabilization is a pre-requisite for Gas Detect mode)	
Gas	3.3.xx
AI Detection Mode	
AI Full Frame Detection	3.8.xx
Custom Classifier	
Custom Classifier	3.2.xx
AI TR Models	3.9.4

2 Detection Modes

This section describes the detection modes that are currently supported.

2.1 Vehicle Moving Target Indication

- Works best finding moving vehicles in a moving scene.
- Works well from a moving platform for finding objects that are typically 10 to 100 pixels long in the scene.
- Not suitable for finding small, large, slow, or erratically moving objects.
- Assists with track designation by detecting up to 10 objects with full tracking, or up to 20 objects when using Advanced Tracking (version 3.9.x and above).



Figure 1: Vehicle Moving Target Indication

2.2 Drone Detection

- Useful for detecting drones in both moving and stationary scenes.
- Best for tracking slow and erratically moving objects.
- Works well from a moving or stationary platform to find drones that are typically 10 to 100 pixels long in the scene.
- Assists with track designation by detecting up to 10 objects with full tracking, or up to 20 objects when using Advanced Tracking (version 3.9.x and above).
- Classification helps reduce operator workload by utilizing state of the art deep learning algorithms to reduce false positives.

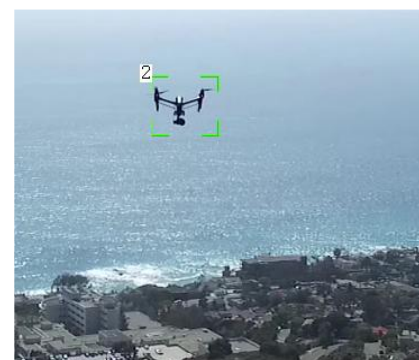


Figure 2: Drone Detection



2.3 Staring Moving Target Indication

- Works best from a ground camera that is either stationary or moves and then stares for an extended time at a scene.
- Detects objects that are small and move slowly, move fast, move at varying rates, or that temporarily stop.
- Detects up to 100 objects.

 *MTD is optional in software 3.11.x and above.*



Figure 3: Staring Moving Target Indication

2.4 Aerial Moving Target Indication (MTI)

- Works best from a moving platform and can find small, slow-moving objects.
- The only moving target mode capable of handling frame-to-frame perspective changes which makes it ideally suited for airborne applications.
- Takes the most computational resources and is likely to impact frame rate (especially on the 1500-OEM).
- Detects up to 100 objects.

 *MTD is optional in software 3.11.x and above.*



Figure 4: Aerial Moving Target Indication (MTI)

2.5 Anomaly Detection

- Works from a moving or stationary platform and finds objects that are unique in the scene based on color and intensity.
- Does not require objects to be moving to be detected.
- Detects up to 100 objects.



Figure 5: Anomaly Detection



2.6 Radiometric Detection

- Works from a moving or stationary platform and finds objects within a specified temperature range.
- Does not require objects to be moving to be detected.
- Detects up to 100 objects.



Figure 6: Radiometric Detection

2.7 Maritime Detection

- Works from a moving or stationary platform and finds objects in a maritime environment such as ships or life jackets.
- Does not require objects to be moving to be detected.
- Detects up to 100 objects.



Figure 7: Maritime Detection

2.8 Blob Detection

- Works from a moving or stationary platform and finds objects that are light, or dark compared to their surroundings.
- Does not require objects to be moving to be detected.
- Detects up to 100 objects.



Figure 8: Blob Detection



2.9 Gas Detection

- Works from a moving or stationary platform.
- Reduces operator workload.
- Enhances and colorizes gases that are visible with specialized cameras sensitive to the methane absorption band.

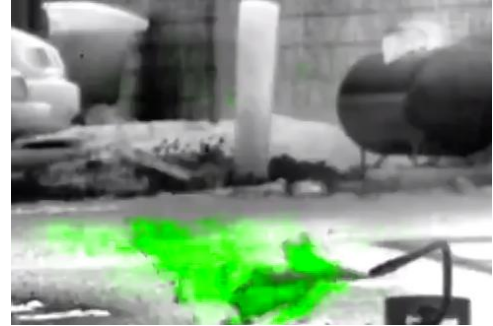


Figure 9: Gas Detection

2.10 Person Detection

- Similar to Vehicle Detection mode but initiates tracking in Person mode, specifically designed for slow and irregularly moving objects such as people.
- Effective from a moving platform for detecting objects approximately 10 to 100 pixels in length within the scene.
- Assists with track designation by detecting up to 10 objects with full tracking, or up to 20 objects when using Advanced Tracking (version 3.9.x and above).



Figure 10: Person Detection

2.11 AI Detection

- Performs full-frame searches utilizing custom deep learning models optimized for modern NPU hardware.
- Effectively detects both stationary and moving objects with models specifically trained for these tasks.
- Simultaneously classifies all detected objects.

MTD is optional in software 3.11.x and above.

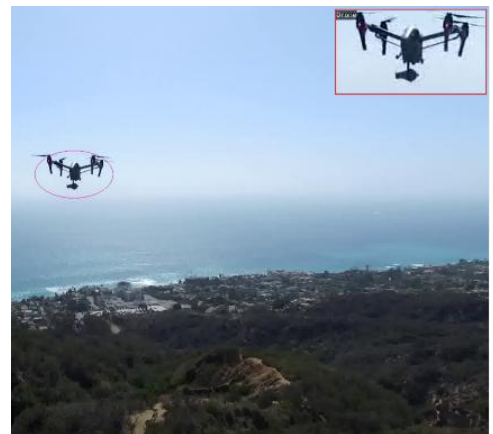


Figure 11: AI Detection



2.12 Motion Detection

- Underlying detection method used for Drone, Vehicle and Person, without requiring track boxes.
- MTD Track boxes are optional to enhance object tracking
- Characteristics of Drone, Vehicle and Person detection modes are available if the corresponding tracking mode is selected with MTD enabled.

 *MTD is optional in software 3.11.x and above.*



Figure 12: Motion Detection

3 Multiple Detections

3.8.x software and above.

Two detections can be run on the same image. Select the second detection by setting the detection *Index* from the Panel Plus main menu » *Track* tab » *Detect* tab. This setting also controls the Advanced Detection parameters and Detection Region of Interest separately for each detection.

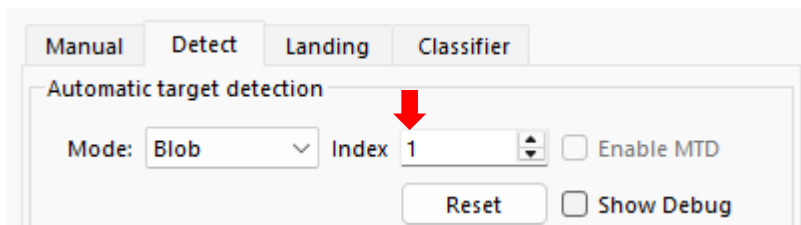


Figure 13: Multiple Detections

3.1 Limitations

Only a single instance of Vehicle, Drone, Person, and Gas detection can run at a time. It must have a detection *Index* of *0*.

Denoise Masking in Enhancement cannot run at the same time as detection with an *Index* of *1*. It can only run with one detection with an *Index* of *0*.

4 Classification

Classification works on detections or track outputs providing the ability to filter detections based on classification labels.

New in 3.5.x software.

Sightline provides pre-trained models that are deployed with the software and selected through Panel Plus. Users can also develop their own trained models. See [EAN-Classifer-Development](#) for more information.



The primary use of the classifier is to filter out detections to reduce false positives. The classifier can be used in combination with the detection mode options. It can also be used without detection filtering. This allows the user to make decisions based on classification labels and confidence values included in the [TrackingPositionsExtended \(0xA0\)](#) message.

The classifier is not available for the 1500-OEM.

4.1 Classifier Settings

Classifier settings can be found in Panel Plus » *Track* tab » *Classifier* tab.

Classifications Max Per Frame: Adjust this number (1 - 10) to increase or decrease the number of times per frame the classifier can run. Each detection is classified approximately three times per second.

Classifications Min Dimensions: Specifies the minimum square dimensions (in pixels) a detection must meet before the classifier attempts to classify the object. This setting is useful for ignoring very small detections that are unlikely to be classified accurately due to insufficient pixel data.

Valid range: 0 - 1000 pixels. Set to 0 to restrict dimension filtering.

Track/detect Filtering: Selected items will not be displayed or show up in telemetry. More details about Classifier Filtering options can be found in the [Classifier Filtering](#) section.

The screenshot shows the 'Classifier' tab in the software interface. It contains several sections: 'Classifier Options' with a checkbox for filtering drone detections; 'Drone Classifier Sensitivity' with a slider from High to Low; 'Tracker Classification' and 'Detection Classification' dropdown menus both set to 'No Tracks' and 'No Detections' respectively; 'Classifications' section with 'Max per frame' set to 1 and 'Min dimensions' set to 0; 'Track/Detection Filtering' section with a 'Filter Configuration' button and a table of filters (SLA ID, Custom ID, and three SubClass Thresholds) all set to 0x0000000000000000; and a 'Custom Classifier File Name' section with a 'Model Loader' button and the file name 'sla_drone.cls'.

Figure 14: Classifier Settings

Custom Classifier File Name:

- **Request Files:** Populates the list of available files on the OEM board. Select an option from the drop-down list or manually enter a name if using the SLA-Library.
- **Update File:** Sends the file name to the OEM board.

See [EAN-Classifier-Development](#) for information on creating custom classifiers and uploading them to the OEM board.



4.2 Sightline Pre-Trained Models

This section provides information about Sightline pre-trained classifier models that are distributed with Sightline software.

While Sightline distributes pre-trained models it does not distribute the data that was used to create these models.

Sightline is constantly working to improve the models and will include new models with future software releases. If the included models are not applicable, a custom classifier can be created and loaded to a supported OEM board to use with any of the detection modes. See [EAN-Classifier-Development](#) for more details.

4.2.1 Drone Model

A drone classification classifier was introduced in 2.25.x software. In 3.5.x, users can select this classifier model (**s1a_drone.cls**) using the *Custom Classifier File Name* feature in Panel Plus. It is compatible with all detection modes but performs best in *Drone* detection mode.

Criteria	Expected Performance
Minimum pixels classification:	10x10
Pixels for more robust classification:	20x20
Performance against <i>In the Wild</i> * data set:	85%
Dataset size:	75000 Drone Image, 75000 Non-Drone Images
Sensor types:	IR and Visible Imagery
Ground or aerial:	Trained against images from ground and aerial sensor platforms.

In 3.8.x software, the classifier model (**s1a_drone_large.cls**) can be selected using the *Custom Classifier File Name* feature in Panel Plus. It is compatible with all detection modes but performs best in *Drone* detection mode.

Criteria	Expected Performance
Minimum pixels classification:	10x10
Pixels for more robust classification:	20x20
Performance against <i>In the Wild</i> * data set:	85%
Dataset size:	96000 Drone Image, 140,000 Non-Drone Images
Sensor types:	IR and Visible Imagery
Ground or aerial:	Trained against images from ground and aerial sensor platforms.

*The *In the Wild* dataset is a carefully selected set of 500 images taken from outside sources to ensure they have not been tested and have not been tested against any images or videos that Sightline has trained on.



4.2.2 Drone MTI Filtering

The *Use Classifier* setting is only available when using the *Drone* detection mode. When enabled this mode prevents detections on tracks with a low probability of being a drone.

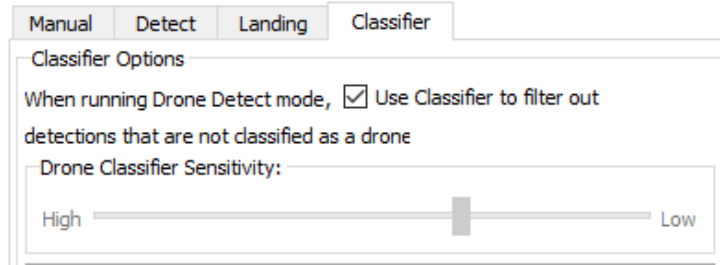


Figure 15: Classifier Options

- In 3.0.xx software the classifier can now be used to filter out detections that are not classified as a drone and/or add a classification label. See the [Classifier Filtering](#) section for more information. Using drone detection without filtering false detections requires the Classifier software feature.
- Drone detection mode is the preferred method for filtering out false detections. This mode combines the probabilities of Rotary Wing Drone and Fixed Wing Drone and filters out tracks where this combined probability is less than a configured threshold based on Drone Classifier Sensitivity.
- [Classifier Filtering](#) and Drone Detection filtering are mutually exclusive; only one can be active at a time.

4.2.3 Vehicle and Person Model

Starting with 3.5.x software a vehicle and person classifier model (`sla_vehicle_person.cls`) is included. This model can be selected using the *Custom Classifier file Name* feature in Panel Plus. It can be used with any detection mode, but is best used with the *Aerial*, *Staring*, *Vehicle*, and *Person* modes.

Criteria	Expected Performance
Minimum pixels classification:	Vehicles: 10x10 People: 16x16
Pixels for a more robust classification:	Vehicles: 20x20 People: 32x32
Performance against In the Wild* data set:	84%
Dataset size:	26000 Vehicle Images, 11000 People Images, 19000 Background Images
Sensor types:	Visible Imagery Only. Future updates may include IR imagery, or a custom classifier could be developed to address this use case.
Ground or aerial:	Trained against images from ground and aerial sensor platforms.

*The In the Wild dataset is a carefully selected set of 500 images taken from outside sources to ensure they have not been tested and have not been tested against any images or videos that Sightline has trained on.



4.3 Classifier Filtering

3.9.x software and below.

Detections can be filtered based on their classification label. For example, in [Figure 16](#) using the `sla_vehicle_person.cls` classifier model, only detections that classified as vehicles would be displayed and included in telemetry.

Classifier Filtering	
☒ Background/Other	☒ Person
☒ Rotary Wing Drone	☐ Custom 0
☒ Fixed Wing Drone	☐ Custom 1
☐ Vehicle	☐ Custom 2

Figure 16: Classifier Filtering

- To use additional Sightline filters or custom filters that are not included in Panel Plus, see the [IDD](#). If a filtering option is checked in Panel Plus all objects classified with the labels being filtered do not show up in telemetry.
- Classifier filtering for Vehicle, Person, and Drone detection modes was added in 3.4.x software.
- When Drone Detection filtering is enabled, Classifier Filtering is overridden.
- Classifier Filtering requires the Classifier software feature.
- Advanced Classifier Filtering settings are available through the [SetSystemValue \(0x92\)](#) message and should be changed with caution.

4.4 Filter Configuration

Track/Detection Filtering	
Filter Configuration	SLA ID Filters: 0X2000000000000000
	Custom ID Filters: 0X0000000000000000
	SubClass Threshold A: 0X0000000000000000
	SubClass Threshold B: 0X0000000000000000
	SubClass Threshold C: 0X0000000000000000

Figure 17: Filter Configuration

4.4.1 Drone Reporting Mode

When running drone detection, the classifier will report either *Drone* or *Background* by default. A more detailed reporting mode can be enabled, and the report will be either *Drone*, *Fixed Wing Drone*, *Background*, or *Vehicle*.

- Prior to 3.5.x software the system only reported detailed results.

4.4.2 Detection Padding

When deciding what area around a detection to classify, additional padding may be needed based on the detection algorithm that is being used. Use the default setting for most cases and adjust as needed.

4.4.3 Update Rate

The confidences and labels shown in telemetry are based on multiple classifications of the target. The rate new samples are incorporated into the output can be adjusted.



4.4.4 KLV Filtering

New in 3.11.x software.

Allows the user to filter targets based on KLV data. Supported filters include minimum and maximum target height and width, minimum and maximum target latitude and longitude, and whether the target is above or below the horizon.

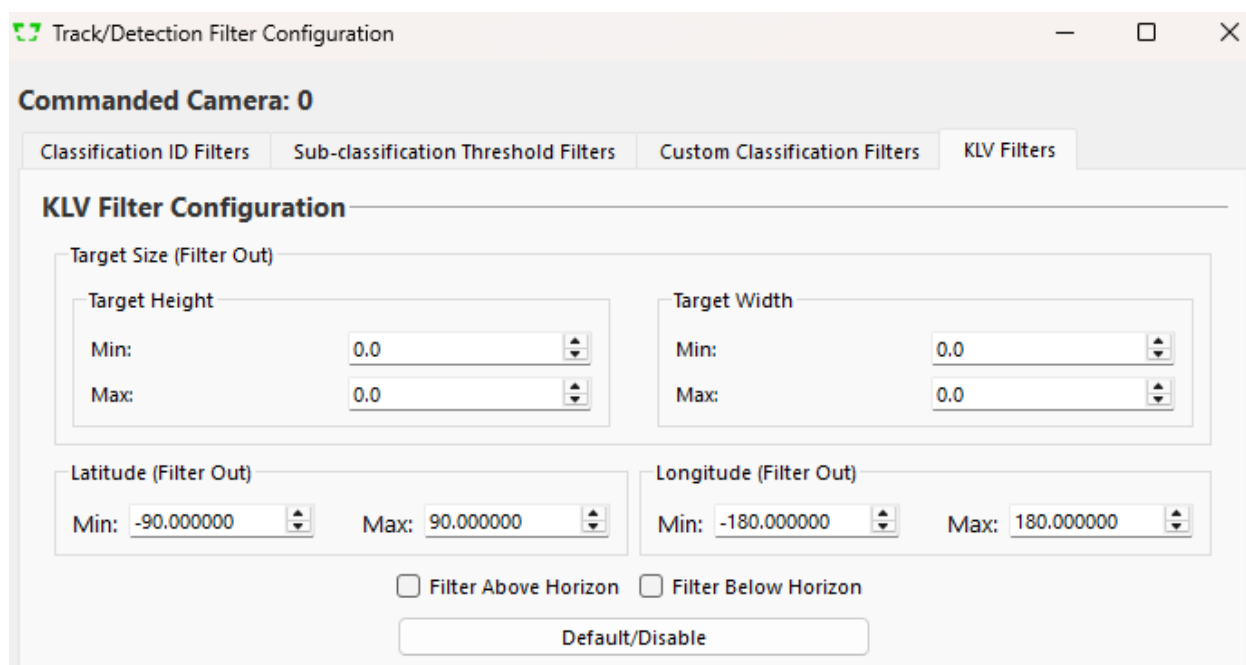
 *Target size minimum and maximum values are specified in meters.*

Target size (height and width) depends on accurate FOV and slant range values from KLV data (see [EAN-KLV-Metadata](#) for more information on FOV and slant range).

Setting minimum and maximum target size values allows automatic filtering of detections that are too large or too small for the defined range. This is useful when the approximate size of the desired target is known, and detections outside that size range should be excluded. Valid values range from 0.0 m to 200.0 m.

The *Above Horizon* and *Below Horizon* filters are useful for automatically removing detections that appear above or below the scene horizon. This is particularly helpful in low-slant, staring scenes to reduce noisy detections that may appear in the sky. These filters require accurate latitude and longitude KLV data.

Latitude and longitude ranges allow the user to define minimum and maximum values, automatically filtering out detections that fall outside the specified geographic bounds.



The screenshot shows the 'Track/Detection Filter Configuration' window with the 'KLV Filters' tab selected. The window title is 'Track/Detection Filter Configuration'. Below the title bar, it says 'Commanded Camera: 0'. There are four tabs: 'Classification ID Filters', 'Sub-classification Threshold Filters', 'Custom Classification Filters', and 'KLV Filters'. The 'KLV Filter Configuration' section contains the following controls:

- Target Size (Filter Out)**
 - Target Height**
 - Min: 0.0
 - Max: 0.0
 - Target Width**
 - Min: 0.0
 - Max: 0.0
- Latitude (Filter Out)**
 - Min: -90.000000
 - Max: 90.000000
- Longitude (Filter Out)**
 - Min: -180.000000
 - Max: 180.000000
- ☐ Filter Above Horizon ☐ Filter Below Horizon
-

Figure 18: KLV Filters Tab



4.5 Managing Compute Resources

The 17xx-OEM and 4100-OEM have a neural processing unit (NPU) that can aid in classification and target recognition. AI Detection models will use the NPU resources and are executed asynchronously from the video stream to ensure a stable frame rate. Classification models can be configured to use either CPU or NPU resources and can be configured to run synchronously or asynchronously with the video stream.

4.5.1 NPU Resources

To force the classifier to use the NPU, a system value has been provided (*NPU_CONTROL*). Setting *Value0* to 1 will force all classification inferencing onto the NPU.

System Value

System Value

41: NPU_CONTROL

☒ Persist (will be saved when parameter file is saved)

Value0 1

Value1 0

Value2 0

Value3 0

Send Request

Figure 19: NPU CONTROL System Value

4.5.2 Asynchronous Execution

By default, classification models are executed synchronously with the video stream. If the model has a high latency, this can cause a drop-in frame rate. To avoid a drop-in frame rate, the classifier can be forced to run asynchronously by configuring the provided system value (*CLASSIFY_ASYNC*). Setting Value 0 to 1 will force all classification inferencing to be executed asynchronously from the video stream.

System Value

System Value

43: CLASSIFY_ASYNC

☒ Persist (will be saved when parameter file is saved)

Value0 1

Value1 0

Value2 0

Value3 0

Send Request

Figure 20: CLASSIFY_ASYNC System Value



5 Detection Configuration Guidelines

Start with the *Detect* default parameters in Panel Plus » *Track* tab » *Detect* tab in the *Mode* drop-down menu shown in [Figure 21](#).

Adjusting *Sensitivity* changes the number of detections (both true and false detections).

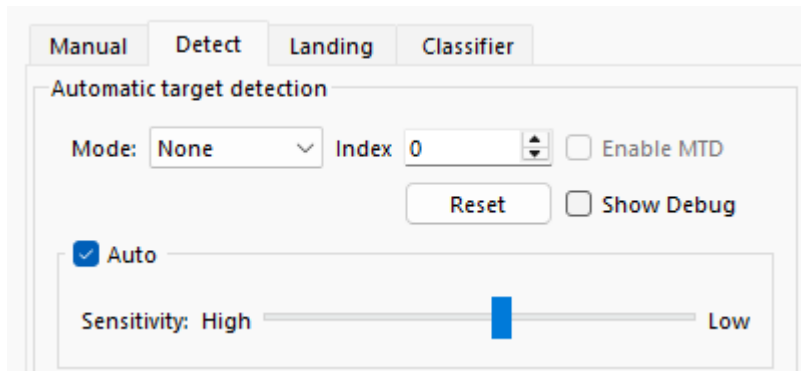


Figure 21: Automatic Target Detection Modes

Advanced detection settings can be found by clicking the *Advanced* button in the *Detect* tab shown in [Figure 22](#).

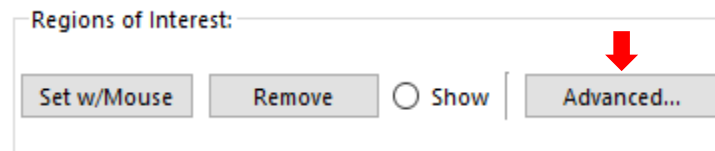


Figure 22: Advanced Detection Settings

The following sections will not describe every parameter in depth but will explain parameters that may need to be adjusted depending on the hardware setup and use case.

Processing at a constant frame rate is important for detection performance. The display processing load can be reduced by turning off enhancement, choosing a faster encoder (MPEG4 for 1500-OEM), and using a smaller output frame size.

The Performance graphs in Panel Plus show the time used by the processing steps. From the main menu View » Performance Graphs.

5.1 Vehicle, Person, and Drone MTI

The primary parameter setting for *Vehicle*, *Person*, and *Drone* detection MTI is *Sensitivity*. Use the default value as a starting point for easy adjustments. Other key settings are *Regions of Interest* and *Downsample* covered in the [Performance and Speed](#) section.

5.1.1 Acquisition Assist

In the [SetTrackingParameters \(0x0C\)](#) message, the *Acquisition Assist* setting, shown in [Figure 23](#), has an impact on *Vehicle*, *Person*, and *Drone* MTI. It uses an algorithm with hints of detection data to make an optimal axis aligned bounding box for the target. If it is not enabled only detection data is used. It also works well if the scene is not cluttered. If the scene becomes too cluttered, it should be disabled.



5.1.2 Intelligent Assist

From the *Manual* tab, enabling *Intelligent Assist* is useful in *Drone* mode. *Intelligent Assist* allows the track box to be resized as needed.

5.1.3 AI Assist

From the *Manual* tab, enabling *AI Assist* allows track boxes to be resized based on the AI model output. This is useful for reinforcing tracks as targets change in size, using an AI detection model output. To use *AI Assist*, the *AI Detection* mode must be enabled.

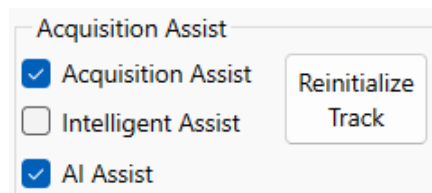


Figure 23: Acquisition, Intelligent, and AI Assist

5.2 Staring MTI

Parameters for this feature can be found by clicking the *Advanced* button in the *Detect* tab. The parameter settings for *Staring MTI* are very similar to *Aerial MTI*. In most cases the default values are sufficient. *Staring MTI* uses two additional parameters

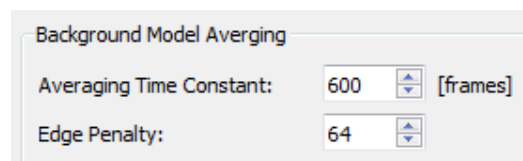


Figure 24: Staring Settings

Average Time Constant: The approximate amount of time to fully replace the background model.

Edge Penalty: Applied to edges in the background model, minimizing minor differences to reduce the number of false detections. The higher the value, the larger the penalty. While lowering the value allows smaller, slower moving objects to be detected, it may increase the number of false detections.

Staring MTI requires the *Stationary Camera Low Drift* setting. This can be found from the Panel Plus main menu » *Configure* » *Stabilization*. This corresponds to setting the *Update Rate* in the [RegistrationParameters \(0x9E\)](#) message to 10.

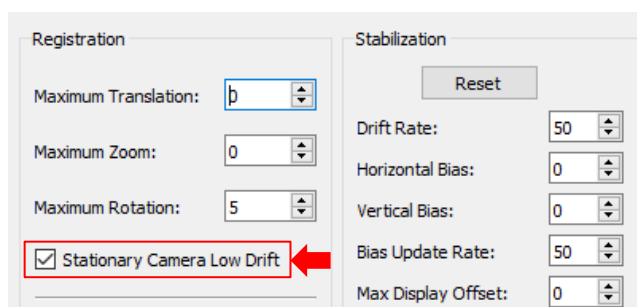


Figure 25: Stationary Camera Low Drift Setting

In software 3.11.x and above MTD can be enabled in *Staring* mode to start tracks on objects. Any track mode can be used with *Staring MTD*.



5.3 Aerial MTI

The *NFramesBack* setting determines how far back in time to search. Reductions in frame rates will need to be changed under certain circumstances. For example, if every other input frame is being skipped, the number may need to be lowered. The number should also be lowered for fast-moving objects (between 6 and 15).

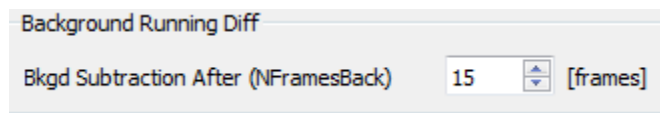


Figure 26: Aerial Settings

 Parameters for this feature can be found by clicking the *Advanced* button in the *Detect* tab.

5.4 Anomaly, Radiometric, Blob, Maritime Detection

Parameters for this feature can be found by clicking the *Advanced* button in the *Detect* tab.

Use Registration: Anomaly, Radiometric, Blob, and Maritime detection modes use this parameter. It is used to control whether detections are tracked from frame-to-frame.

 When looking for detections on each frame independent of other frames, i.e., when scanning a large area, disable this setting.

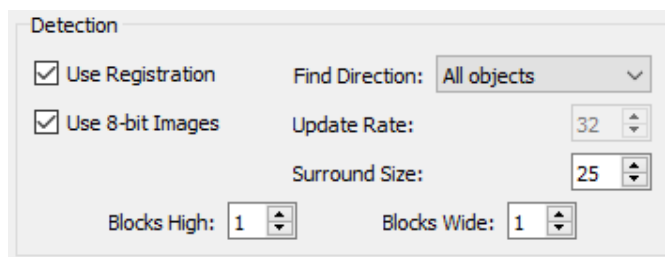


Figure 27: Anomaly, Radiometric, Maritime, Blob, Detection Settings

 When slowly scanning or staring, use the registration results along with a higher number of watch frames to reduce the number of false detections.

Update Rate: Anomaly and Maritime use this parameter. It controls how fast the detection model is updated. Larger values put more emphasis on the current frame.

Typical values are in the 8 to 64 range. For example, when running *Maritime* detection mode in high seas, lowering this number helps reduce the number of false detections. When running in calm seas where registration may be less accurate, increasing this value can be beneficial.

Surround Size: Determines how large an area around each pixel to consider in the blob detection algorithm. This number should be larger than the size of the object that is being detected. If detections appear only on one side of the object or only the outline of the object in the debug image, the number should be increased.

Find Direction: Select bright or dark objects, or both bright and dark objects to find (all objects). New in 3.5.x software. Prior versions of software only found both bright and dark objects (all objects). This feature only applies to the *Blob* detection mode.

Use 8-bit Images: Use 8-bit images for detection rather than 16-bit images to get faster processing times. New in 3.6.x software. This feature only applies to *Blob*, *Radiometric*, and *Anomaly* detection modes.



In software 3.11.x and above MTD can be enabled in Aerial mode to start tracks on objects. Any track mode can be used with Aerial MTD.

The default value for Surround Size is 25 starting with software version 3.5.x, which is a good starting point. The previous default was 7 in software version 3.4.x and below.

5.5 Radiometric Detection

Radiometric detection shown in the *Detect* tab looks for pixels that fall within a specified range of gray values. This algorithm can operate on 8-, 14-, or 16-bit data.

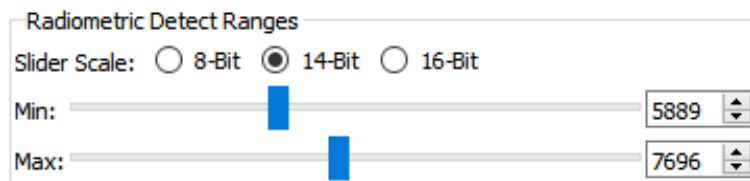


Figure 28: Radiometric Detect - Temperature Mode

16-bit mode operation available in 3.2.x software.

5.6 Gas Detection

Parameters for this feature can be found by clicking the *Advanced* button in the *Detect* tab.

Use the settings in *Background Model Averaging* to control the discrimination of gas features from the background.

Averaging Time Constant: The approximate amount of time in frames to fully replace the background model. For gas, a small value between 5 and 20 is recommended. Larger values can make the measurement more sensitive, but only for stationary cameras. Smaller values work better for cameras with low shake or small movements.

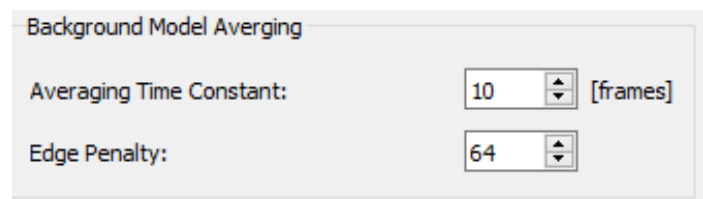


Figure 29: Gas Detection Settings

Edge Penalty: Applied to edges in the background model, minimizing minor differences to reduce the number of false detections. The higher the value the larger the penalty. A value of 64 is recommended for gas detection.

Use the settings in *Gas Enhance* to control the displayed enhancement of gas.

Color: Select the color of the enhanced gas. Select *White* for no color.

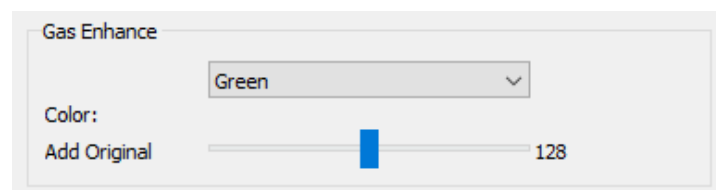


Figure 30: Change Gas Overlay Color and Parameters

Add Original: Controls the fraction of the original image that is shown with the gas differences. 0 will only show gas differences. 128 will show 50% of the original image intensity plus the gas differences. 255 will show 100% of the original image plus the gas differences.



8/16-Bit: Choose *16-Bit* to allow the algorithm to run on the original 16-bit image (before auto gain) if it is available. Choose *8-Bit* to always execute on the image after it has gone through auto gain. *16-Bit* allows the detection to remain sensitive even in regions that get saturated by auto gain.

5.7 Motion Detection

In the *Frame Depth* section, the *Number of Frames* controls the number of frames in the past that the algorithm uses to find and detect motion. Longer frames pick up smaller or harder to detect objects. Shorter frame depth is best used for faster moving objects.

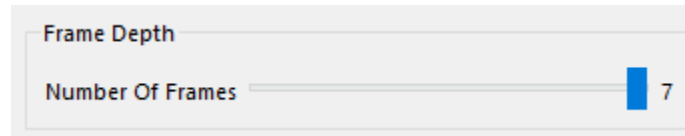


Figure 31: Number Of Frames Past

The *Motion* detection mode is the underlying detection used for *Vehicle*, *Drone* and *Person* detection modes. However, this mode does not require MTD tracks (optional) to allow users to use more object-based detections and tracking as optional.

5.8 AI Detection

AI Detection: Adjusts the intersection over union (IoU) threshold for AI detection boxes. Lower values merge detection boxes with smaller overlaps, while higher values require a greater overlap before merging. This setting is useful for distinguishing closely positioned or overlapping objects in the detection process.

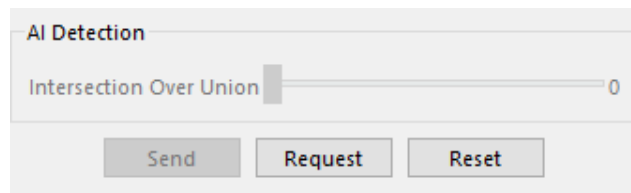


Figure 32: AI Detection IOU

AI Detection uses full frame object detection to find objects based on an AI model. The included trained AI models (*cuas_XXX.slaod*) on the 17xx-OEM and 4100-OEM are trained to identify medium to large fixed-wing and rotary drones. Other included AI models (*person_vehicle_XXX.slaod*) are trained to identify vehicles and people.

Custom AI models can be trained and run on the SLA hardware with support from Sightline. Contact [Sightline Support](#) for more information.



5.8.1 Model Selection Guidelines

The included trained AI models specify the model input size and, optionally, the model architecture size in the file name. The model input size is indicated as 360, 540, 720, or 1080 representing the height of an input image with a 9:16 aspect ratio. The model architecture size is denoted as either **n** (nano) or **s** (small).

Selecting the appropriate model for an application depends on the captured image size and the required frame rate. When the captured image is larger than the input size of the model, it is downsampled using the smallest whole-number factor that fits within the input dimensions of the model.

For optimal performance, choose a model with an input resolution that is an integer factor of the captured resolution. For example, if the captured resolution is 1920×1080, both a 540 model and a 720 model will downsample by a factor of 2. In this case, both models produce the same results, but the 540 model has lower latency.

The table below outlines the expected frames per second (FPS) for each combination of model input size and model architecture on the 4100-OEM, 17xx-OEM, Jetson Orin NX, and Jetson Xavier NX.

Table 3: Average Expected FPS by Model Input Size and Architecture Across Hardware Platforms

Model Input Size (WxH)	Model Architecture Size	4100-OEM FPS	17xx-OEM FPS	Orin NX (4-Core 10W) FPS	Orin NX (8-Core 25W) FPS	Xavier NX (4-Core 10W) FPS	Xavier NX (6-Core 20W) FPS
360 (640x360)	n	88	36	-	-	-	-
540 (960x540)	n	39	16	45	120+	50	77
720 (1280x720)	n	20	-	28	95	43	52
1080 (1920x1080)	n	-	-	14	50	22	27
360 (640x360)	s	45	-	-	-	-	-
540 (960x540)	s	15	-	-	-	-	-
720 (1280x720)	s	-	-	13	47	19	25
1080 (1920x1080)	s	-	-	-	25	-	12

NVIDIA power modes display maximum allowed wattage, not average power consumption.

5.8.2 AI-MTD

When *Enable MTD* is checked in AI detection mode, tracks are automatically created from the detection output. The track behavior depends on the active track settings and operates independently of the detection model that is loaded.

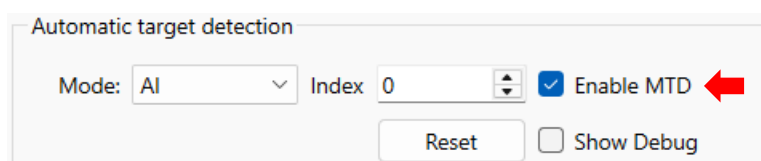


Figure 33: Enable AI-MTD

This mode is useful for tracking targets in challenging scenarios where detection output may be less reliable, such as maintaining tracks through obfuscations.

The behavior is similar to other MTD detection modes (*Drone*, *Vehicle*, *Person*). However, unlike those modes, which require use of their corresponding track mode (for example, *Drone* detect mode requires *Drone* track mode), AI detection output functions independently of track settings. This independence enables more refined track behavior across a wider range of scenarios.



In software 3.11.x and above this MTD can be enabled in Aerial, Staring, Motion and AI Detection modes.

For more information on track modes and track settings, see [EAN-Target Tracking](#).

5.9 Auto and Manual Sensitivity

In the [SetDetectionParameters \(0x2D\)](#) message for Aerial, Staring, Anomaly, Maritime, Blob, Gas, Radiometric and AI detection modes, the sensitivity can be controlled using the Auto or Manual settings in the Automatic target detection section in the Detect tab.

Auto: Easiest to use. The sensitivity directly controls other parameters. A higher sensitivity will result in a greater likelihood of detecting a target but has a greater likelihood of false detections.

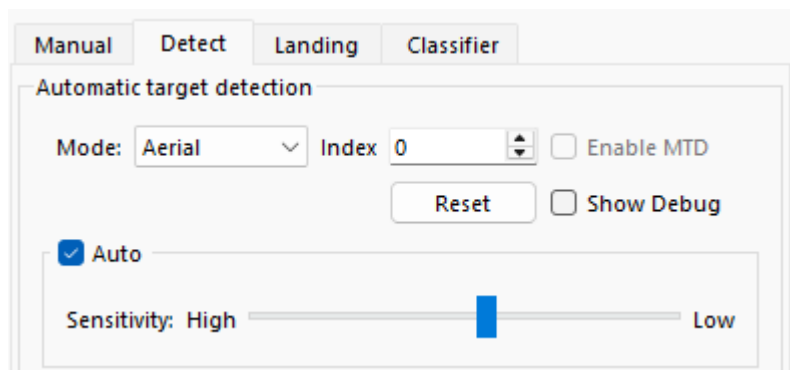


Figure 34: Auto Sensitivity

Manual: Allows the user to directly control the background threshold, watch frames, and suspicious score. The sensitivity setting does not apply.

- **Bkgd Threshold:** Defines how different an object needs to be from the background model to be detected. In the example shown in [Figure 35](#), a low background threshold number will generate more detections.
- **Watch Frames:** Defines how long a target should be watched before reporting it to the user. A lower number of watch frames will result in shorter detection times and can result in more false positives.
- **Suspicious Score:** Defines a level at which the algorithm will consider a track to be unstable based on a history of the track detections.



Figure 35: Manual Sensitivity



Example:

A leaf that blows in the wind moves erratically and would require a high suspicious score to show up as a detection. A car driving in a straight line would require a much lower suspicious score to show up as a detection.

 *The suspicious score does not apply to the Anomaly, Radiometric, Blob, Gas, or Maritime detection modes. The background threshold parameter does not apply to the Radiometric detection mode.*

5.10 Detection Overlap and Tracks

In the [SetAdvancedDetectionParameters \(0x76\)](#) command, this section covers *hideOverlapTrks* and *detNearTrk*. The controls are located in the *Advanced Detection* settings under *Multiple Objects*. These two settings are mutually exclusive.

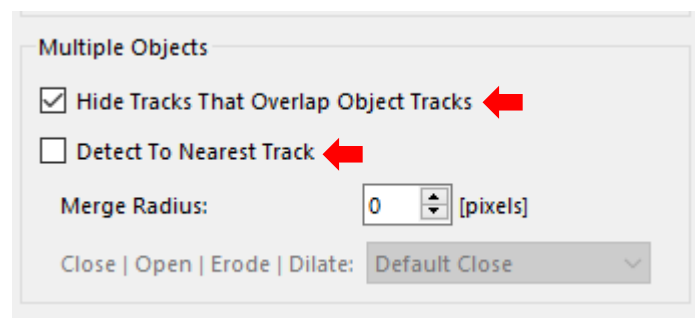


Figure 36: Detection Overlap and Track Controls

Hide Tracks That Overlap Object Tracks: Enabled by default. This stops MTI detections from appearing under Tracks. Disabling this feature allows *Detections and Tracks* to appear on top of each other.

Detect to Nearest Track: Suppresses AI MTI detections with a 50% or greater IOU. The suppressed classification data is then associated with the track overlapping the detection. This setting can only be enabled if *Hide Tracks That Overlap Object Tracks* is disabled.

6 Performance and Speed

This section reviews the tradeoffs between performance (how well the algorithm does at detecting objects) and speed (how long the algorithm takes to run). Settings are in Panel Plus » *Track tab* » *Detect tab* » *Advanced*.

6.1 Downsampling

 *Parameters for this feature can be found by clicking the Advanced button in the Detect tab.*

In *Auto* mode, (shown in [Figure 37](#)) the downsampling rate is automatically set and is based on the image size and algorithm.

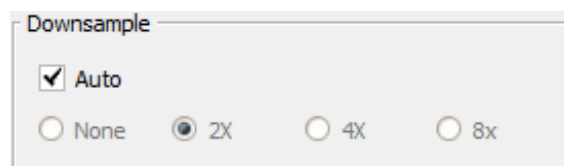


Figure 37: Downsampling Options



Increasing the downsampling time allows bigger and faster targets to be detected (faster processing time). Lowering the downsampling time allows for slower and smaller moving targets to be detected (slower processing time).

Downsampling can also be adjusted to detect slower and smaller targets with HD inputs. This may reduce the output frame rate or require the user to change other parameters such as the Region of Interest to maintain full frame rate.

Downsampling does not apply to anomaly detection.

6.2 Regions of Interest

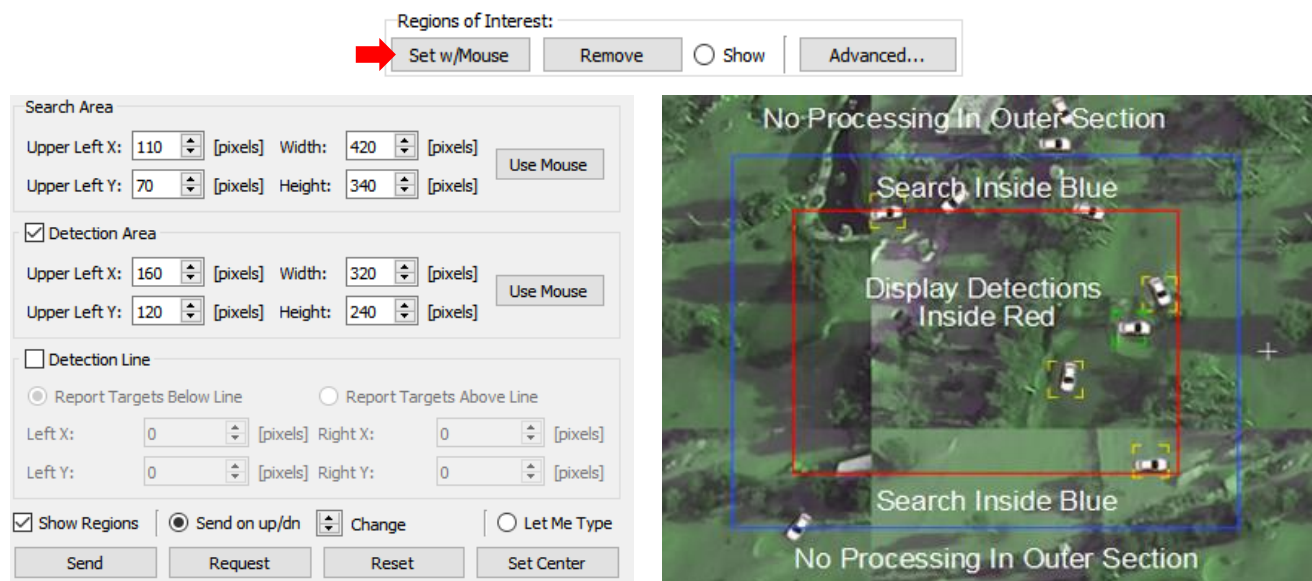


Figure 38: Regions of Interest Example

Regions of Interest can be used to control the speed and performance of the various detection algorithms. In the example below the search area is shown in blue. This region directly impacts the speed of the algorithm since this is the region that will be processed.

The detection area is shown in red in the example below. Detections will be reported in this area. Make the search region slightly larger to detect objects as soon as they enter the detection region. When using the *Set Mouse* option from Panel Plus, the detection and search areas are set to the same size.

Both areas can be the same size, but the detection area cannot be larger than the search area.

IMPORTANT: Since adjusting the search region causes the detect algorithm to be reset it should not be changed rapidly. The detection region or detection line can be changed without resetting the algorithm.



6.2.1 Detection Lines

New in 3.5.x software.

A line can also be used to define a detection region as shown in **Figure 39**. In this case detections can be specified to be reported above and below the line.

The line is defined by two points with the *Right X* point always being larger than the *Left X* point.

 *The software treats the line as extending across the entire search region even if the points are not on the edges of the imager or search region.*

Search Area

Upper Left X: 110 [pixels] Width: 420 [pixels] Use Mouse

Upper Left Y: 70 [pixels] Height: 340 [pixels]

☐ Detection Area

Upper Left X: 0 [pixels] Width: 0 [pixels] Use Mouse

Upper Left Y: 0 [pixels] Height: 0 [pixels]

☒ Detection Line

☒ Report Targets Below Line ☐ Report Targets Above Line

Left X: 0 [pixels] Right X: 640 [pixels]

Left Y: 480 [pixels] Right Y: 0 [pixels]

☒ Show Regions ☐ Send on up/dn ☒ Let Me Type



Figure 39: Regions of Interest Example

6.2.2 Masked Detection Regions

New in 3.6.x software.

Search Area

Upper Left X: 0 [pixels] Width: 0 [pixels] Use Mouse

Upper Left Y: 0 [pixels] Height: 0 [pixels]

☒ Detection Area

Upper Left X: 0 [pixels] Width: 0 [pixels] Use Mouse

Upper Left Y: 0 [pixels] Height: 0 [pixels]

☐ Detection Line

☒ Report Targets Below Line ☐ Report Targets Above Line

Left X: 0 [pixels] Right X: 0 [pixels]

Left Y: 0 [pixels] Right Y: 0 [pixels]

Detection Mask

Blocks High: 8 Blocks Wide: 8

Mask 0: f0f0f0 Mask 1: 0

Mask 2: 0 Mask 3: 0

☒ Enable Detection Mask

☒ Show Regions ☐ Send on up/dn ☒ Let Me Type



Figure 40: Masked Detection Regions



Block-based masks are used to define specific regions to ignore during detection. While this setting does not have as significant an impact on overall processing time as other region of interest settings, it allows for masking out regions at a more precise level.

To use this setting first set the desired number of blocks per row (*Blocks Wide*) and per column (*Blocks High*). Then, use the four hexadecimal masks to specify which blocks should be masked-out from detection. Each mask controls 64 blocks. For example, *Mask 0* controls the first 64 blocks, with each subsequent mask controlling the next 64 blocks.

Check *Show Regions* to display the blocks on the screen. Red blocks indicate masked-out regions and blue blocks indicate unmasked regions.

The masked regions can be used in conjunction with the search region. If a search region is specified, the blocks will be confined within that region. Otherwise, they will fill the entire source image.

Masked regions do not apply to Vehicle, Drone, or Person modes.

6.3 Frame Step

Frame stepping for detection is no longer supported in version 3.3.x software and above. All previous uses of detection frame stepping will default to a value of 0.

If frame stepping is still required, use the frame step found in the capture settings.

6.4 Bit Depth

By default, some detection modes will automatically use 16-bit images if the input image is also 16-bit. To speed up processing times, select the *Use 8-bit Images* option for detection. However, this may potentially impact the detection performance. Note that this option does not affect the bit depth of the output image.

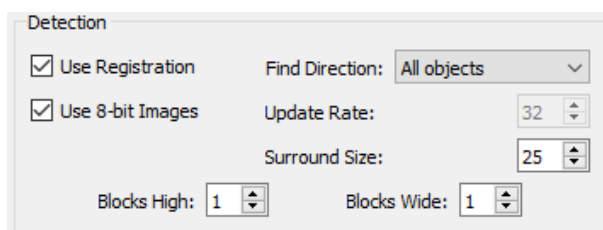


Figure 41: Bit Depth Location

This is only applicable when using detection modes Anomaly, Radiometric, and Blob.



7 Filtering Detections

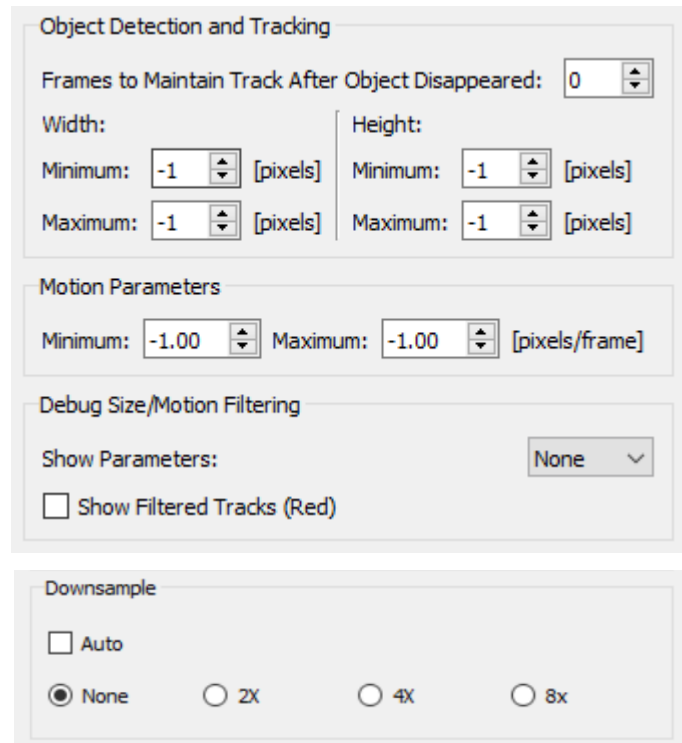
7.1 Size and Velocity Filtering

Detections can be filtered based on different parameters including size and velocity. Settings are in Panel Plus » *Track tab* » *Detect tab* » *Advanced*. Setting these parameters correctly can help filter out false detections.

 *Special care should be taken when searching for very small detections, down to 2x2 pixels, especially when the Downsample level does not allow for these smaller detections. For the smallest size detection (2x2), a Downsample level of None is required.*

The *Debug Size/Motion Filtering* parameters can be used to help understand current values as well as what is getting filtered out.

 *Motion-based filtering does not apply to Vehicle, Drone, Person, or Gas detection. Size-based filtering does not apply to Gas detection.*



The screenshot shows the 'Object Detection and Tracking' settings panel. It includes a 'Frames to Maintain Track After Object Disappeared' slider set to 0. Below this are 'Width' and 'Height' settings, each with 'Minimum' and 'Maximum' spinners set to -1 [pixels]. The 'Motion Parameters' section has 'Minimum' and 'Maximum' spinners set to -1.00 [pixels/frame]. The 'Debug Size/Motion Filtering' section has a 'Show Parameters' dropdown set to 'None' and an unchecked checkbox for 'Show Filtered Tracks (Red)'. At the bottom, the 'Downsample' section has an unchecked 'Auto' checkbox and four radio buttons: 'None' (selected), '2X', '4X', and '8X'.

Figure 42: Filtering Settings

7.2 Classification Filtering

The classifier can be used to reduce the number of false positives. When an object is detected in the scene, the classifier will evaluate that object. Detections can then be filtered based on the detection mode and classifier settings. See the [Classifier Filtering](#) section for additional information on classification.

8 Snapshots of Detections

New in 3.2.x software.

Snapshots of detections can be used for varied reasons depending on the application. For example, to reduce the amount of data sent over a network or to collect data for training a classifier. See the *Configure Detection Snapshots* section in the [EAN-File-Recording](#) for more information on this feature.



9 Command and Control

The following packets are used to set up and control the detection algorithms. Details on these can be found in the [IDD](#).

9.1 Reporting Results

[TrackingPosition \(0x43\)](#)

[TrackingPositions \(0x51\)](#)

[TrackingPositionsExtended \(0xA0\)](#)

[CoordinateReportingMode \(0x0B\)](#)

[SetTelemetryDestination \(0x64\)](#)

9.2 Display Related Commands

[SetOverlayMode \(0x06\)](#)

[SetDisplayParameters \(0x16\)](#)

[DoDetectSnapshot \(0xAB\)](#)

9.3 Detection Related Commands

[SetDetectionParameters \(0x2D\)](#)

[SetAdvancedDetectionParameters \(0x76\)](#)

[SetDetectionRegionofInterest \(0x7C\)](#)

[SetTrackingParameters \(0x0C\)](#)

10 Troubleshooting

Issue	Recommendation
Expected detections are not shown.	Enable the debug window. This shows what the algorithm is looking at.
Additional troubleshooting.	Turn off all detection modes, overlays, and stabilization. Record 1 to 2 minutes of video and send it to Sightline. Sightline will post-process the video, provide an analysis of performance, and then make parameter recommendations.

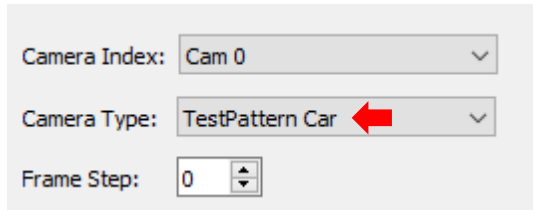
10.1 Questions and Additional Support

For questions and additional support, please contact [Sightline Support](#). Additional support documentation and Engineering Application Notes (EANs) can be found on the [Product Support Documentation](#) page in the [Sightline Knowledge Base](#).

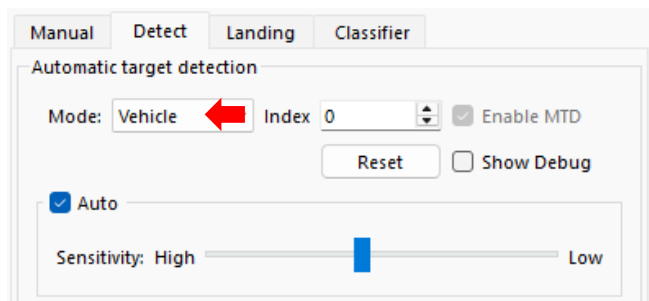


Appendix A - Testing MTI

1. From the main menu in Panel Plus, go to *Configure » Acquisition Settings*.
2. Under the Camera Type drop-down menu, select *TestPattern Car* or *TestPattern Multi Car*.
3. The *Apply* button will turn red indicating input field changes have been detected. Click *Apply*.



4. From the main menu, go to *Parameters » Save to board*. Cycle the system power or go to *Reset » Board*.
5. After the board resets, the test pattern video should be displayed in Panel Plus. Go to *Track* tab » *Detect* tab. In the *Mode* drop-down menu select *Vehicle*.



6. The system should detect the car and place a detection box around it as shown in **Figure A1**.



Figure A1: Detection Box

The mouse can be used to designate other targets.

7. To exit the test pattern, from the main menu in Panel Plus, go to *Configure » Acquisition Settings*.
8. Under the *Camera Type* drop-down menu, select a previous camera setting or select *None*.
9. Click *Apply*.
10. From the main menu, go to *Parameters » Save to board*. Cycle the system power or go to *Reset » Board*.



Appendix B - MTI Overlays

This section gives an overview on how to enable detection overlays. For more detailed information refer to [EAN-Overlay-Graphics](#).

Advanced control of these graphical objects can be found in the *Overlay Settings* dialog window in Panel Plus as shown in **Figure B1**. From the Panel Plus main menu » *Configure* » *Overlays*.

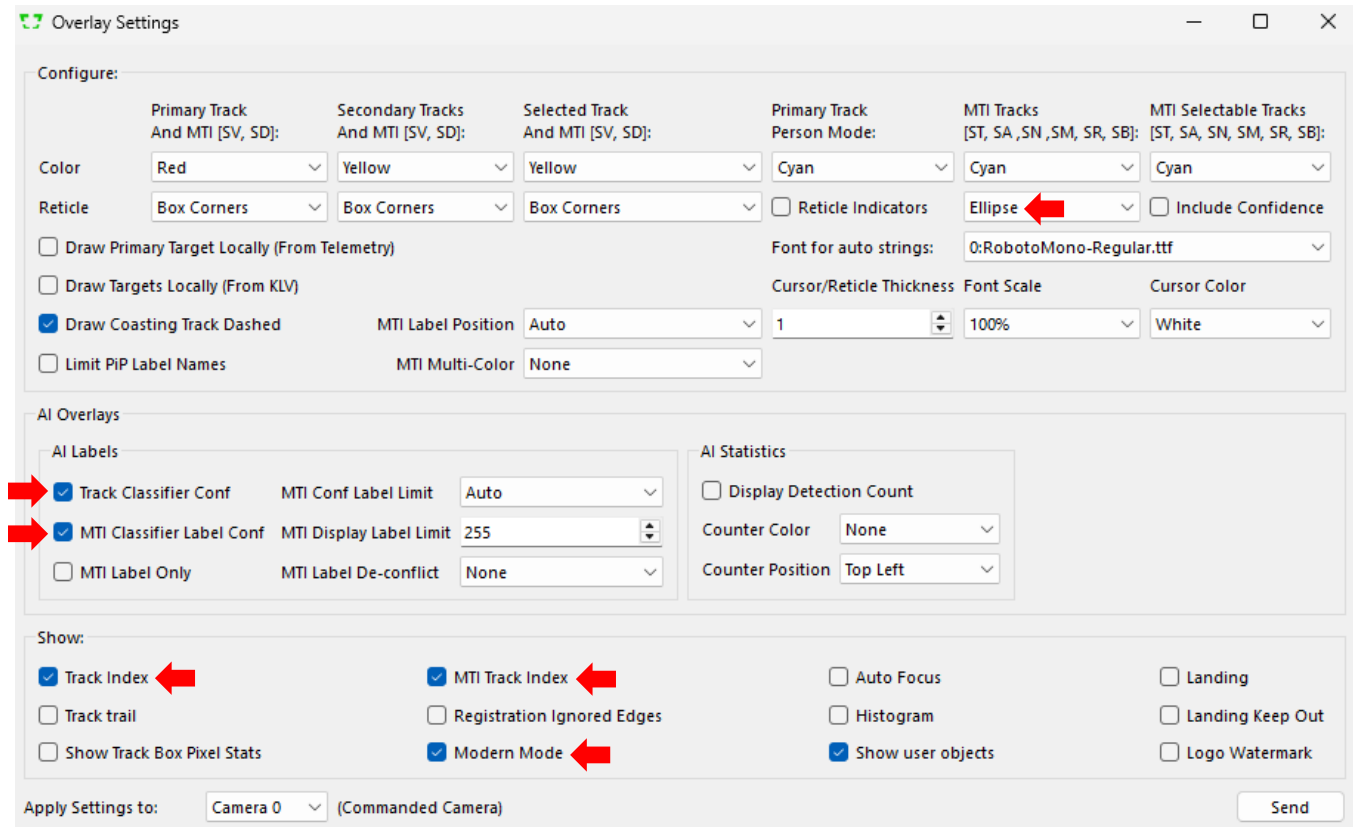


Figure B1: Advanced Overlay Settings Dialog

For MTI detections the most useful overlay graphics to enable are *Track Index* and *Track Classifier Conf* (for Person, Vehicle, Drone). For all other detection modes use *MTI Track Index* and *MTI Classifier Label Conf*.

Primary, secondary and MTI track colors can be changed using the *Color* dropdown menus associated with each track type.

Track Classifier Conf and MTI Classifier Label Conf: Assigns a classification label and confidence score to tracks and MTI tracks. This feature is particularly useful when users want to avoid a PiP window obscuring the displayed video. The confidence values reflect either classifier confidences or AI Detection Mode confidences. See [EAN-TF-Classifer-Development](#) for setup and configuration.

Modern Mode: Enables updated fonts and modern overlay graphics.



MTI Reticle: This drop-down menu allows selection of different MTI detection reticle shapes. Options available include *Ellipse*, *Ellipse + Letter*, *Triangle*, *Triangle + Letter*, *Rectangle*, and *Rectangle + Letter*. The default shape is *Ellipse*, which is suitable for most detection modes as shown in [Figure B2](#).

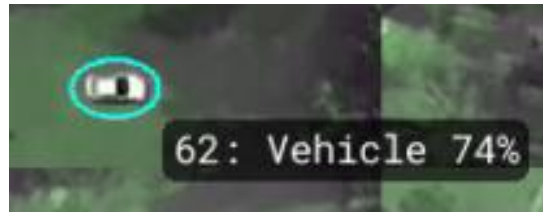


Figure B2: Ellipse Example

When AI detection and/or classification is enabled, using the *+ Letter* options can be beneficial. The letter will be the first character of the classification of each object. For example, a *Vehicle* classification will display a **V** above the reticle.

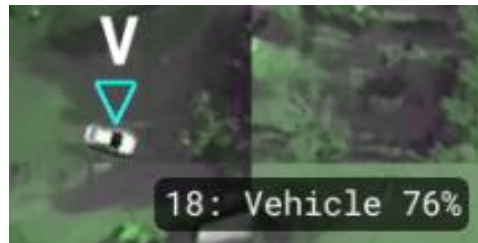


Figure B3: Triangle + Letter Example

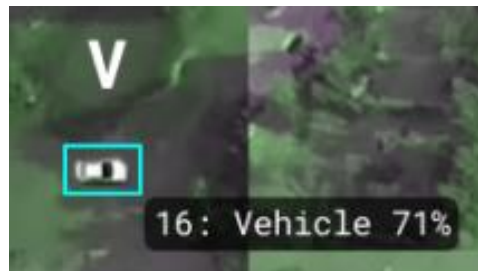


Figure B4: Rectangle + Letter Example